



École Polytechnique

PRL IN COMPUTER SCIENCE

Quadrangulation Sandbox

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Academic year 2024/2025

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1. Introduction

Quadrangulations are important objects in combinatorics at the bridge of discrete mathematics and geometry. This study aims to explore quadrangulations, from their definitions and properties to their bijective connections with trees and visual realization as embedded surfaces.

During the semester, my supervisor Professor Bettinelli taught me most of the quadrangulation theory I got to know. From there, I was free to dig deeper and explore the applications discussed in this report.

It was like he gave me tools to build with. The title “Quadrangulation Sandbox” stems from this.

2. Quadrangulation

2.1. Definitions

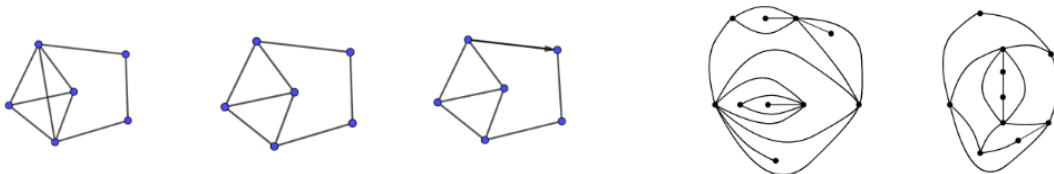
Before we start working on anything, we need to establish clear definitions.

We will be working in a planar space.

Definition (1): A planar map is an embedding of a connected planar graph in the plane with adjacent faces, non-crossing edges and cyclic order of edges.

Definition (2): A rooted map is a planar map with a distinguished, oriented edge called the root edge.

Definition (3): A quadrangulation is a planar map where every face has degree 4.



Non-planar map, Planar map, Rooted map

Quadrangulations (J. BELTRAN, J.-F. LE GALL) [2][12]

2.2. Counting

We now take interest in counting the number of rooted quadrangulations with n faces.

Euler's Formula (4): For any connected planar graph:

$$V - E + F = 2$$

With V number of vertices, E number of edges, F number of faces, g the genus of the surface ($g=0$ for planes or spheres).

Property (5): By (4), a quadrangulation with n faces has $2n$ edges and $n+2$ vertices.

Proof: In a quadrangulation, the sum of face degrees is $4n$. Since each edge is adjacent to two faces:

$$2E = 4n \Rightarrow E = 2n$$

Applying Euler's formula (4):

$$V - 2n + n = 2 \Rightarrow V = n + 2$$

□

Definition (5): Catalan numbers are a classic sequence in combinatorics given by:

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

Here we dive into a very short digression on Catalan numbers [3][4]:

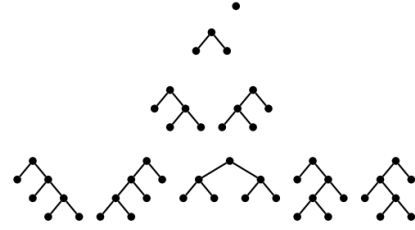
The Catalan numbers satisfy the recurrence:

$$C_0 = 1, \quad C_{n+1} = \sum_{i=0}^n C_i \cdot C_{n-i}$$

and appear in many combinatorial decompositions such as:

()	C1 = 1
() (), (())	C2 = 2
() () (), () (()), (())(), (()) (), ((()))	C3 = 5
()()()(), ()()()(), ()()()(), ()()()(), ()()()(), ()()()(), ()()()(), ()()()(), ()()()(), ()()()(), ((()))(), ((()))(), ((()))(), ((()))()	C4 = 14

Grouping parentheses around sub-expressions



Splitting a binary tree at its root (T. DAVIS)

Tutte's Formula (6): The formula for the number of rooted planar quadrangulations with n faces is given by:

$$q_n = \frac{2 \cdot 3^n \cdot (2n)!}{n! \cdot (n+2)!} = \frac{2 \cdot 3^n}{(n+2)(n+1)} \cdot C_n$$

2.3. Cycles

We're also interested in counting the length of cycles.

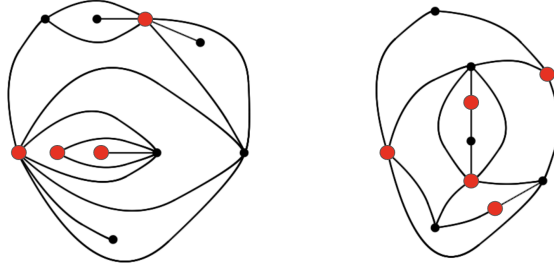
Property (7): Quadrangulations have cycles of even length.

Proof: Assume that there exists a cycle of odd length in a quadrangulation.

Definition (3) implies that the underlying graph is bipartite: starting from any vertex, we can 2-color the vertices, alternating colors as we move across edges. The graph remains properly 2-colored throughout.

Yet a cycle of odd-length would return to the starting vertex with the opposite color $\Rightarrow$ contradiction.

□



Quadrangulations (J. BELTRAN, J.-F. LE GALL) with added bipartite coloring

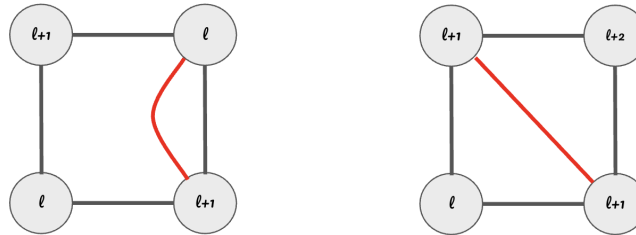
3. Schaeffer Bijection

With established definitions, we're all set to start exploring.

The first instance of it will be through the Schaeffer bijection. Indeed, quadrangulation maps we've studied so far are nice and interesting, but a tree can for example enable facilitated enumeration, encoding, and algorithmic applications. [1][9]

3.1. From map to tree

Given a rooted planar quadrangulation with n faces, we assign to each vertex v a label $\ell(v)$, equal to its graph distance from the root vertex. Since quadrangulations are bipartite (7), labels differ by exactly 1 across adjacent vertices. For each face, we draw a new edge (red) depending on the labels as follows:



How to draw edges

The red edges over all faces form a plane tree with its neighboring vertices differing in label by at most 1.

3.2. From tree to map

Given a well-labeled plane tree with n edges and root label 0, we introduce a new node v^* outside the tree, with $\ell(v^*) = 0$. For each node of the tree with label ℓ , connect it to the next node with label $\ell-1$ in counterclockwise order, wrapping cyclically. The added edges form the dual edges of a quadrangulation. They do not cross due to the tree's plane embedding.

3.3. Proof of the bijection

From map to tree:

We want to prove that the graph T obtained from the mapping is a tree.

By (4) and the description of the mapping, we know that going from a planar map with n faces and $n+2$ vertices, we obtain a tree with n edges and $n+1$ vertices. T has the correct number of edges and vertices for a tree and is connected, so it is sufficient to show that it is acyclic.

Assume that T contains a cycle C . Since red edges are only introduced by joining distinct vertices of adjacent labels, cycles cannot be of length 1. Moreover, a cycle of length 2 requires 2 red edges within the same face, so it is also impossible. So C has length at least 3. Let $a \in C$ be the vertex of minimal label along the cycle, and let $\ell = \ell(a)$. Let b and c be the neighbors of a in the cycle. Then: $\ell(b), \ell(c) \in \{\ell, \ell+1\}$.

We then have two cases:

- b has label ℓ and c has label $\ell+1$ (or the opposite). The edge (b, a) implies there is a vertex of label $\ell-1$ connected to b outside C . The edge (c, a) implies another vertex of label $\ell-1$ connected to c inside C .
- $\ell(b) = \ell(c) = \ell+1$. Then the pair (a, b) gives a red edge to a vertex of label $\ell-1$ outside C , and (a, c) gives a red edge to a vertex of label $\ell-1$ inside C .

In either case, we have two distinct vertices of label $\ell-1$ on opposite sides of C .

Since all labels are derived from distances to the root vertex, any path from it to both of these $\ell-1$ vertices crosses C . But this is impossible under a minimal distance labelling.

So C cannot exist. T is acyclic, connected and well-labeled. So it is a tree. □

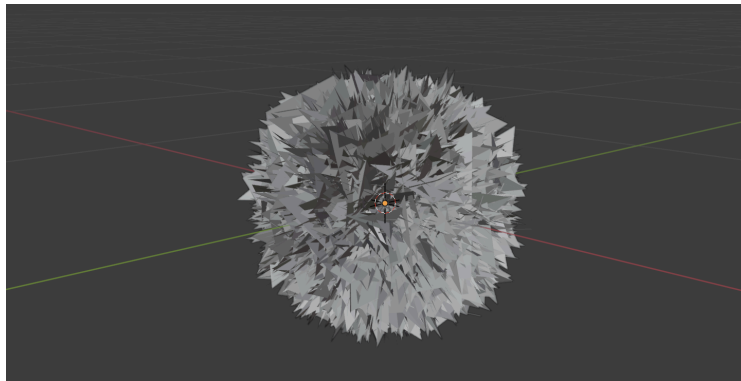
From tree to map:

We want to prove that the graph M obtained from the mapping of a well-labelled plane tree T is a map.

- Assume M is not planar (some edges intersect). Then take nodes a, b, c, d where a, c and b, d are connected. By construction we have $\ell(a) = \ell(c) + 1$ and $\ell(b) = \ell(d) + 1$. Since edges are formed to the nearest corner with label 1 lower, we expect: $\ell(b) \geq \ell(a)$ and $\ell(c) \geq \ell(b)$. But $\ell(c) \geq \ell(b) \geq \ell(a) = \ell(c) + 1$ is a contradiction. So M is planar
- For any pair of adjacent vertices in the tree, there exists a corresponding path in the quadrangulation, so M is connected.
- Each red edge added between ℓ and $\ell-1$ forms part of a face composed of the root vertex, 2 adjacent nodes, and the endpoint of the red edge $\ell-1$. Thus we get a closed face formed by four vertices and four edges. So every face in M has degree 4.

□

3.4. Visualization



Random planar quadrangulation (10)

To better understand how the algorithm works, I practiced making a visualization of a large random planar quadrangulation.

We begin by generating a random well-labeled plane tree with n edges, where labels encode graph distance to the root (see 2.2). It is then converted into a planar quadrangulation using the bijection (see 2.3). The construction was automated using the `simquad` software [7], which implements this bijection to generate uniform random rooted planar quadrangulations with a specified number of faces (10 000 in this case).

The combinatorial map output was interpreted as a graph and embedded in three-dimensional space using a spring layout algorithm: vertices are arranged by imagining that each edge is a spring pulling connected points together, while all points also push away from each other. The final layout balances these forces to spread out the structure clearly in space [6]. The resulting geometry was rendered in Blender (see 10 above).

We notice that even though the quadrangulation starts out as a list of connections, once we lay it out using the spring layout, the overall shape it forms looks similar to how large random maps behave when you embed them. For further discussions we can mention that for large sizes, these embeddings start to resemble a mathematical object called the Brownian map [8].

Through this visualization, I got to see a glimpse of the complex geometry that appears in the scaling limit.

3.5. Discussion

We have covered the Cori-Vauquelin-Schaeffer bijection. These other bijections offer natural directions for future study, though they lie beyond the boundaries of our sandbox.

	Plane Surface	Orientable Surface	Non-Orientable Surface
Quadrangulation	Cori-Vauquelin-Schaeffer	Chapuy-Marcus-Schaeffer	Chapuy-Dolega
General Maps	Bouttier-DiFrancesco-Guitter	Combination of the 2 adjacent	Bettinelli

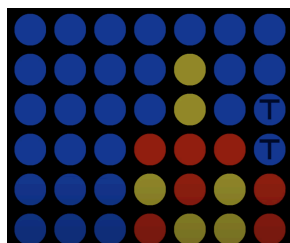
4. Connect 4

Around halfway through the semester, I was on a call with my little cousin and was explaining my PRL subject to him. He did not understand it nor find it as interesting as I did. As a way to remedy this, I started trying to relate it to something that he'd like hearing about: Connect 4, a game we play together a lot. With the work done on quadrangulations and bijections, this seemed like an interesting exploration point.

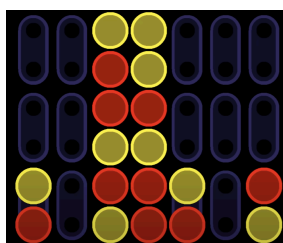
4.1. Rules and strategies

As is widely known, Connect 4 is a two-player game where players take turns dropping colored discs into a 7×6 grid. The first player to align four of their discs vertically, horizontally, or diagonally wins.

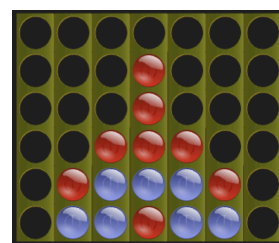
Due to the popularity of this game, it has been widely researched. Giving rise to numerous winning strategies [11], examples being:



Threat Pairing
Chess equivalent of “fork”



Clameven
Forcing your opp. to open threats on odd rows



Septuple Check
Red has 7 ways to win

4.2. Winning

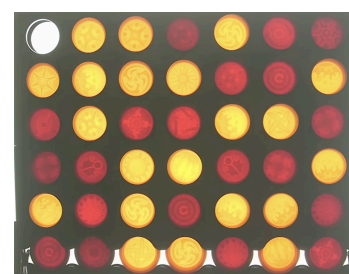
Definition (8): A solved game is one where the outcome can be correctly predicted from any position, assuming both players play optimally.

Property (9): Connect 4 is a solved game [15].

Proof:

With a blank, there are over 4 trillion possible game states. To reduce our computation needs, we first eliminate all symmetric plays, which leaves us with over 2 trillion possible games. We also use alpha-beta pruning to eliminate from the get-go branches that wouldn't affect the final decision. Then, we compute the solution by modeling the game as a decision tree. The root is the blank board and from then each node represents a possible board state. We explore the tree with the Minimax algorithm, going through it by evaluating the best possible move for each player at each turn.

□



“The perfect game”

4.3. Quadrangulations

Given the subject of this project, a large amount of time was dedicated to trying to find a bijection between a Connect 4 board and quadrangulations. The 2 lead ideas were:

- We model the board as an initially empty planar graph, where each move adds a vertex corresponding to a placed disc. Edges are then defined based on local adjacency (horizontal, vertical, diagonal) to form a subgraph. The hypothesis is that a player wins if their induced subgraph contains a quadrangulation. But in most play-throughs, adding vertices does not preserve planarity; and if players create sparse patterns, we also fall short of connectivity. This model thus proves to not quite work for quadrangulation verification.
- We model the board as a fixed planar graph, where each cell is a vertex and edges connect adjacent cells (horizontally, vertically, and diagonally). A game position is a partial vertex coloring of this graph, and a move consists of coloring an uncolored vertex according to the game's rules. The hypothesis is that a player wins if the pattern of their colored vertices forms a quadrangulated region on the board. While this model keeps planarity by construction, it fails to align with the actual winning condition of the game. In particular, players win by forming linear chains of four tokens, which are path-like and do not correspond to enclosed, face-bounded structures as required by quadrangulations. Furthermore, the set of a player's occupied cells typically does not induce a subgraph where all interior faces have degree four. So this model also proves to not quite work for quadrangulation verification.

Discussion: we can speculate that winning positions could be associated to quadrangulations through a bijection defined in a different combinatorial or geometric space. Alternatively and more broadly, the space of possible boards could be encoded as a tree or graph whose regions or transitions correspond to quadrangulations under some transformation. Connect 4 is an age-old game that has been heavily dissected so I did not expect to find anything new, but this still furthered my knowledge and was a valuable thought experiment.

4.4. Extra fun

To better grasp optimized connect 4, I took some time to build my own bot to play against.

To be able to play games more or less live (around 10 seconds of computation at every move), I set up a depth limit to the algorithm. Below are some games I played (and lost) against it.

[]	[]	[]	[0]	[]	[]	[]
[]	[]	[X]	[0]	[X]	[0]	[]
[]	[]	[X]	[0]	[0]	[X]	[X]
[]	[]	[0]	[X]	[0]	[X]	[X]
[]	[]	[X]	[0]	[X]	[0]	[X]
[0]	[]	[X]	[0]	[0]	[0]	[X]
1	2	3	4	5	6	7

[]	[]	[]	[0]	[]	[]	[]
[]	[]	[]	[X]	[]	[]	[]
[]	[]	[X]	[X]	[X]	[X]	[]
[]	[]	[0]	[X]	[0]	[0]	[]
[]	[]	[0]	[0]	[X]	[0]	[X]
[]	[]	[X]	[0]	[0]	[0]	[X]
1	2	3	4	5	6	7

[]	[X]	[0]	[0]	[X]	[]	[]
[]	[X]	[X]	[0]	[0]	[]	[]
[]	[0]	[0]	[X]	[X]	[]	[]
[]	[X]	[0]	[0]	[X]	[]	[]
[]	[0]	[0]	[X]	[0]	[0]	[]
[X]	[X]	[X]	[0]	[0]	[X]	[X]
1	2	3	4	5	6	7

[]	[]	[]	[]	[]	[]	[]
[]	[]	[]	[]	[]	[]	[]
[]	[]	[]	[]	[]	[]	[]
[]	[]	[]	[]	[]	[]	[]
[]	[]	[]	[]	[]	[]	[]
[]	[]	[]	[0]	[]	[]	[]
[]	[]	[]	[X]	[]	[]	[]
1	2	3	4	5	6	7

wait a min...
DEPTH 10
Suggested move: Column 4
Enter your's move (1-7): █

The next logical step was to write a program that would tell me what to play, an assistant. Same as before, I had to set up a depth limit and it is thus not perfect because it looks for the best move only up to a certain point. Nevertheless this was good enough to take on my cousin.

5. Conclusion

This study aimed to discover what quadrangulations are and how they can be constructed, encoded, and visualized; thereby deepening my graph theory and combinatorics understanding, particularly on planar maps, tree bijections, and recursive decompositions.

My work also included applying quadrangulation ideas to model Connect 4. While this exploratory connection does not end up producing a direct bijection, it helped me grasp the boundaries between structure and strategy, and provided a playful counterpoint to the more formal mathematical content.

I also got super good at the game, which contributed to making this a valuable experience.

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